Red-light traffic signal with ultrasonic “camera” (NET3001 Final)

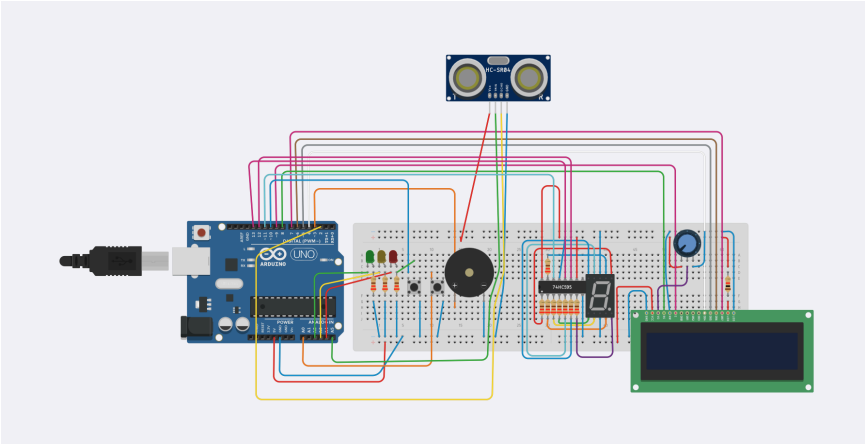
Name of the group members and contributions

- Abdul Rehman (Student 1):
  - TinkerCAD
  - Video
  - serial testing
- Isaac Abdile (Student 2):
  - Hardware
  - Firmware, LCD/7-seg
  - report

Brief description of your system

The traffic light works as follows. 10 s green, 5 s amber, 10 s red. LCD shows phase text; 7-segment countdown via 74HC595. Buttons pause/resume and reset to green at 10. On red, ultrasonic detects entering near zone; buzzer beeps once; LCD shows VIOLATION; USART prints VIOLATION dist=XX cm.

Circuit diagram



List of components and topics used

Components:

- Uno
- 3 LEDs
- Resistors
- 2 buttons
- LCD
- 7-seg + 595
- HC-SR04
- Buzzer
- Breadboard

Detailed explanation (pseudo-code)

```
INIT pins, LCD, 7-seg, USART, Timer1 1 Hz, PCINT on buttons.
phase=GREEN, seconds=10, running=TRUE.
LOOP: handle debounced pause/reset; on 1 s tick update violation message timer or countdown phase;
    if RED and running, read ultrasonic, detect far-to-near while armed, trigger violation and USART;
    re-arm when far again.
```

Short reflection

Hardest part: tuning zone thresholds and keeping ISRs short. Next time: USART calibration mode; optional Timer1 input capture on echo.